

TASK 1 – RTL DESIGN & FUNCTIONAL SIMULATION

8-Bit ALU using SystemVerilog

Digital VLSI Internship Assignment Report

Project	8-Bit ALU RTL Design and Functional Simulation
HDL	SystemVerilog
Tool	Xilinx Vivado 2025.1 / XSim
Verification	Directed + 100 random tests
Total Tests	110
PASS	110
FAIL	0

1. Objective

The objective is to design a synthesizable 8-bit Arithmetic Logic Unit (ALU) in SystemVerilog and verify its functional behavior using a testbench containing directed and random tests. The testbench checks the result, zero flag, and carry flag against expected values.

2. ALU Operations

Opcode	Operation	Description
000	ADD	A + B with carry-out
001	SUB	A - B; carry indicates A >= B
010	AND	Bitwise AND
011	OR	Bitwise OR
100	XOR	Bitwise XOR
101	NOT	Bitwise inversion of A
110	LEFT SHIFT	A shifted left by one; carry = A[7]
111	RIGHT SHIFT	A shifted right by one; carry = A[0]

3. RTL Design

The design is implemented in the module **alu_8bit**. It accepts two 8-bit operands A and B and a 3-bit opcode, and produces an 8-bit result with zero and carry status outputs. A 9-bit temporary value is used for addition so the carry-out can be retained.

RTL source: **alu_8bit.sv**

4. Functional Verification

The testbench instantiates the ALU and uses a reference model for independent checking. It performs directed tests and 100 random tests. Every test compares DUT outputs with the expected result and reports PASS or FAIL.

5. Simulation Procedure

The RTL and testbench were simulated in Vivado XSim. The simulation was run until all tests completed. Console output was checked for the final verification status and waveform viewing was used to inspect signal activity.

6. Verification Results

Metric	Result
Total tests	110
PASS	110
FAIL	0
Operations covered	All 8
Overall status	ALL TESTS PASSED

7. Test Coverage

All eight operation codes were exercised. Recorded counts were: 000 = 16, 001 = 15, 010 = 13, 011 = 12, 100 = 12, 101 = 16, 110 = 16, and 111 = 10. Thus every supported ALU operation was covered during functional verification.

8. Evidence / Screenshots

Figure 1 – Functional simulation / console result

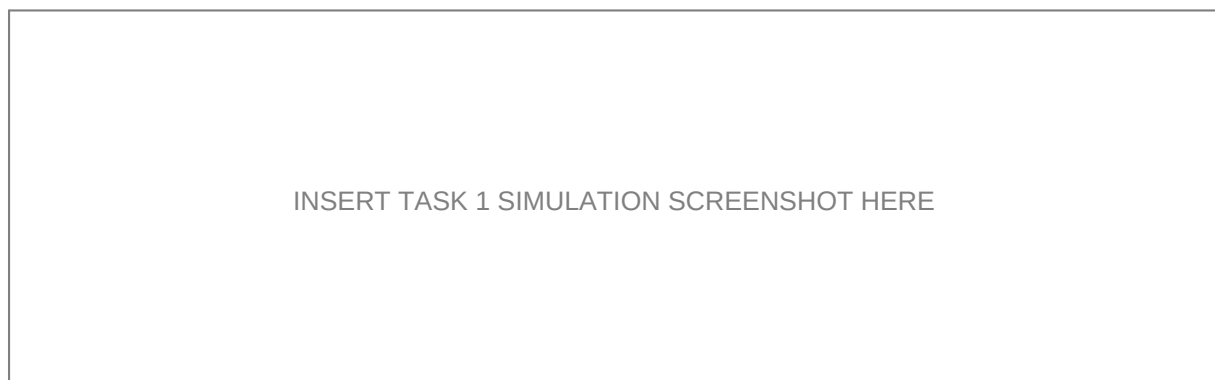


Figure 2 – Simulation waveform

INSERT TASK 1 WAVEFORM SCREENSHOT HERE

9. Conclusion

The 8-bit ALU was functionally verified using SystemVerilog and Vivado XSim. A total of 110 tests were executed, with 110 passes and zero failures. All eight supported operations were exercised. The task demonstrates the RTL design and functional simulation flow for a small digital VLSI block.

10. Deliverables

- alu_8bit.sv – synthesizable RTL
- alu_8bit_tb.sv – verification testbench
- Functional simulation evidence
- Waveform evidence
- This report

Submission note: Replace the two screenshot placeholders with the actual screenshots before final submission if your evaluator requires the images embedded inside the PDF.